CUSTOMER CHURN PREDICTION PROJECT

Final Machine Learning Project Documentation

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1. Introduction

Customer Churn Prediction is one of the most important real-world applications of Machine Learning. The main objective of this project is to predict whether a customer is likely to leave a company or continue using its services.

Many industries such as telecom, banking, insurance, streaming platforms, and e-commerce companies use churn prediction systems to reduce customer loss and improve business growth.

This project applies multiple Machine Learning algorithms to analyze customer behavior and identify patterns responsible for customer churn.

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2. Problem Statement

Companies lose revenue when customers stop using their services. Identifying customers who are likely to leave helps businesses take preventive actions such as:

- Personalized offers
- Discounts
- Customer support improvements
- Retention campaigns

The goal of this project is:

To build a Machine Learning model that predicts customer churn based on customer information and usage patterns.

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3. Objectives of the Project

The major objectives of this project are:

- Understand customer behavior using data
- Perform data preprocessing
- Build classification models
- Compare multiple Machine Learning algorithms
- Evaluate model performance
- Select the best-performing model
- Predict churn for new customers

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4. Technologies Used

Technology	Purpose
Python	Programming language
Pandas	Data handling

Technology	Purpose
NumPy	Numerical operations
Matplotlib	Visualization
Scikit-learn	Machine Learning
Jupyter Notebook	Development environment

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5. Machine Learning Concepts Covered

This project covers multiple important Machine Learning concepts:

Topic	Usage
Classification	Predict churn
Logistic Regression	Baseline model
Decision Tree	Tree-based prediction
Random Forest	Ensemble learning
KNN	Distance-based classification
Train-Test Split	Model evaluation
Confusion Matrix	Performance analysis
Precision / Recall / F1-score	Advanced evaluation
Cross Validation	Stable evaluation
Hyperparameter Tuning	Model optimization
Feature Importance	Important feature analysis

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6. Dataset Description

The dataset contains customer-related information such as:

Feature	Description
Age	Customer age
MonthlyCharges	Monthly bill amount
Tenure	Number of years with company
Contract	Subscription type
InternetService	Internet connection type
SupportCalls	Number of support complaints
Churn	Target variable

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7. Understanding the Target Variable

The target column is:

Churn

Value	Meaning
1	Customer will leave
0	Customer will stay

Since the output contains categories, this becomes a:

Classification Problem

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8. Data Preprocessing

Data preprocessing is one of the most important steps in Machine Learning because real-world data may contain:

- Missing values
- Categorical data
- Irrelevant features
- Inconsistent data formats

The preprocessing steps performed in this project are:

8.1 Handling Missing Values

Missing values reduce model quality. They are handled using methods such as:

- Mean replacement
- Median replacement
- Mode replacement

8.2 Label Encoding

Machine Learning models require numerical input. Therefore categorical values are converted into numerical form.

Example:

Contract Type	Encoded Value
Monthly	0
Yearly	1

8.3 Feature Selection

Relevant columns are selected as input features.

Input Features (X)

- Age
- MonthlyCharges
- Tenure
- Contract
- InternetService
- SupportCalls

Target Variable (y)

- Churn

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9. Train-Test Split

The dataset is divided into:

Data Type	Purpose
Training Data	Used to train model
Testing Data	Used to evaluate model

Generally:

- 80% → Training
- 20% → Testing

This helps evaluate model performance on unseen data.

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10. Machine Learning Models Used

Multiple classification algorithms are used in this project.

10.1 Logistic Regression

Logistic Regression is a supervised classification algorithm used for binary classification problems.

It predicts probability values between 0 and 1.

Advantages

- Simple
- Fast
- Easy to interpret

10.2 Decision Tree

Decision Tree works using rule-based splitting.

It creates a tree structure:

- Root node
- Decision nodes
- Leaf nodes

Advantages

- Easy visualization
- Handles non-linear patterns

Limitation

- Can overfit

10.3 Random Forest

Random Forest is an ensemble learning algorithm that combines multiple decision trees.

It improves:

- Accuracy
- Stability
- Generalization

Advantages

- Reduces overfitting
- High performance
- Feature importance support

10.4 K-Nearest Neighbors (KNN)

KNN predicts class based on nearest neighboring data points.

Advantages

- Simple
- Easy to understand

Limitation

- Slow on large datasets

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11. Model Evaluation

Model evaluation is essential to determine how well the model performs.

11.1 Accuracy Score

Accuracy measures the percentage of correct predictions.

Formula concept:

$$Accuracy = \frac{Correct\ Predictions}{Total\ Predictions}$$

Higher accuracy generally indicates better performance.

11.2 Confusion Matrix

Confusion Matrix helps analyze prediction quality in detail.

It contains:

Term	Meaning
TP	True Positive
TN	True Negative
FP	False Positive
FN	False Negative

Confusion Matrix provides deeper insight than accuracy alone.

11.3 Precision

Precision measures how many predicted positive cases are actually positive.

11.4 Recall

Recall measures how many actual positive cases are correctly predicted.

11.5 F1-Score

F1-score balances:

- Precision
- Recall

It is very useful for imbalanced datasets.

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12. Cross Validation

Cross Validation improves evaluation reliability.

Instead of using only one train-test split, the dataset is divided into multiple folds.

Benefits:

- More stable performance
- Better model comparison
- Reduced bias

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13. Hyperparameter Tuning

Hyperparameter tuning improves model performance by selecting the best parameter values.

Examples:

- max_depth
- n_estimators
- n_neighbors

GridSearchCV is used to find the best parameter combination.

14. Feature Importance

Feature Importance identifies which features contribute most to prediction.

Example:

Feature	Importance
MonthlyCharges	High
Tenure	Medium
Age	Low

This helps understand customer behavior better.

15. Model Comparison

Different models are compared based on:

- Accuracy
- Precision
- Recall
- F1-score

Example comparison:

Model	Accuracy
Logistic Regression	82%
Decision Tree	85%
Random Forest	91%
KNN	84%

16. Best Model Selection

After comparing all models:

Random Forest was selected as the best model

because:

- highest accuracy
- lower overfitting
- stable predictions
- good generalization

17. Final Prediction

The final trained model predicts whether a new customer is likely to churn.

Example:

- Customer age
- Monthly bill
- Tenure
- Support calls

Using these inputs, the model predicts:

0 → Stay
1 → Leave

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18. Real-World Applications

Customer Churn Prediction is widely used in:

- Telecom companies
- Banking systems
- OTT platforms
- Insurance companies
- Subscription services
- E-commerce businesses

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19. Challenges Faced

Some common challenges in churn prediction include:

- Imbalanced datasets
- Missing values
- Feature selection
- Overfitting
- Hyperparameter tuning

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20. Future Improvements

Future improvements can include:

- Deep Learning models
- Larger datasets
- Real-time prediction systems
- Deployment using Flask/Django
- Cloud integration

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21. Learning Outcomes

This project helps understand:

- Complete Machine Learning workflow
- Data preprocessing
- Classification algorithms
- Evaluation techniques
- Model optimization
- Real-world prediction systems

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22. Conclusion

Customer Churn Prediction is an important Machine Learning application that helps companies reduce customer loss and improve retention strategies.

In this project:

- multiple Machine Learning models were implemented,
- evaluated using various metrics,
- and compared to identify the best-performing algorithm.

Random Forest achieved the best performance and provided accurate churn prediction results.

This project demonstrates the practical implementation of Machine Learning concepts in solving real-world business problems.